CCIT Report #571

January 2006

On Subband Transform Coding for Color Image Compression

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Abstract

Although subband transform coding is a useful approach to image compression, the performance of this method has not been analyzed so far for color images, especially when the selection of color components is considered. Obviously, the RGB components are not suitable for such a compression method due to their high inter-color correlation. On the other hand, the common selection of YUV or YIQ is rather arbitrary and in most cases not optimal.

In this work we introduce a rate-distortion model for color image compression and employ it to find the optimal color components and optimal bit allocation (optimal rates) for the compression. We show that the DCT (Discrete Cosine Transform) can be used to transform the RGB components into an efficient set of color components suitable for subband coding. The optimal rates can be also used to design adaptive quantization tables in the coding stage with results superior to fixed quantization tables. Based on the presented results, our conclusion is that the new approach can improve presently available methods for color compression.

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